

ADR IDENTIFICATION BASICS

Task 1 | Case-Based Pharmacovigilance Project

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Task type	Fictional educational training case
Submission format	GitHub repository + PDF case report

Project objective: demonstrate a structured approach to recognizing a suspected ADR, reviewing the timeline, considering alternative explanations, and documenting the reasoning in a submission-ready format.

CONFIDENTIALITY NOTE

This is a fictional educational case created for internship learning. It is not a real patient record and should not be used for clinical decision-making.

1. Pharmacovigilance Foundations

Pharmacovigilance (PV) focuses on the detection, assessment, understanding and prevention of adverse effects and other medicine-related problems. Continuous safety monitoring remains important because some uncommon, delayed or population-specific reactions become clearer only after wider medicine use.

AE vs ADR - the working distinction

Feature	Adverse Event (AE)	Adverse Drug Reaction (ADR)
Core idea	An untoward medical occurrence during treatment; causality does not need to be established.	A noxious, unintended response associated with a medicine used at normally intended doses.
Causality	Not necessarily caused by the medicine.	A suspected or assessed relationship with the medicine is considered.
Practical question	"What happened?"	"What happened, and is a medicine suspected to be responsible?"

ADR pattern overview

Type	Pattern	Typical feature	Training example
A	Augmented	Predictable; related to known pharmacology; often dose-related.	Hypoglycaemia with insulin.
B	Bizarre	Generally unpredictable; can involve hypersensitivity or idiosyncratic responses.	Penicillin allergy.
C	Chronic	Linked to long-term use.	Adrenal suppression with prolonged corticosteroid exposure.
D	Delayed	Appears after a delay.	Some drug-related cancers or teratogenic effects.
E	End-of-use	Associated with withdrawal.	Withdrawal symptoms after abrupt discontinuation of certain medicines.
F	Failure	Unexpected failure of therapy.	Reduced contraceptive effectiveness due to an interacting medicine.

2. Fictional ADR Identification Case

Case title: Acute urticarial reaction following diclofenac initiation

Patient: 29-year-old man with intermittent mechanical back pain.

Background medicines: Pantoprazole 40 mg once daily for approximately four months.

New medicine: Diclofenac 50 mg twice daily started for acute back pain.

Day 2: The patient develops an itchy, raised red rash over the trunk and upper limbs with mild eyelid puffiness.

Exposure review: No new food, cosmetic product, insect exposure, antibiotic or other newly started medicine is reported.

Action: Diclofenac is stopped. Symptoms begin to settle within 24 hours and are almost completely resolved by Day 4.

Rechallenge: Not performed.

Medicine timeline

Day	Exposure / event	Safety relevance
Day -120 to -1	Pantoprazole continued without reported reaction.	Long-standing concomitant medicine; not the leading temporal suspect.
Day 0	Diclofenac 50 mg twice daily started.	New exposure establishes the primary temporal anchor.
Day 2	Pruritic urticaria + mild eyelid puffiness appear.	Reaction follows recent initiation.
Day 2	Diclofenac discontinued.	Begins the dechallenge observation window.
Day 3-4	Rash and puffiness progressively resolve.	Improvement after withdrawal supports a drug-event relationship.

Initial assessment

Element	Assessment
Reaction	Generalized pruritic urticarial rash with mild eyelid puffiness.
Suspected medicine	Diclofenac - newly initiated immediately before onset.
Temporal relationship	Compatible: reaction appeared about 48 hours after initiation.
Dechallenge	Positive: symptoms improved after diclofenac withdrawal.
Rechallenge	Not performed, so no re-exposure evidence is available.
Alternative causes	No obvious new food, cosmetic, insect exposure or other newly started medicine reported.

3. Causality Reasoning

Causality assessment asks how likely it is that the medicine caused the observed reaction. For this fictional case, the reasoning below uses the same factors highlighted in the task material: timing, dechallenge, rechallenge, alternative explanations and known reaction patterns.

NEW EXPOSURE	~48 H	POSITIVE	NONE
diclofenac started	time to onset	dechallenge	rechallenge data

WHO-UMC-style assessment snapshot

Factor	Case evidence	Interpretation
Reasonable time relationship	Reaction began about two days after starting diclofenac.	Supports a temporal association.
Alternative explanations	No newly introduced food, cosmetic, insect exposure or medicine identified in the case.	No clear competing trigger documented.
Dechallenge	Symptoms eased after diclofenac was stopped.	Supports the suspected relationship.
Rechallenge	Not performed.	No additional confirmation available.
Previous information / pharmacology	The presentation is compatible with a hypersensitivity-type pattern.	Adds context, but does not independently establish causality.

Preliminary conclusion: The event is consistent with a potential diclofenac-associated hypersensitivity-type ADR. The timing and positive dechallenge support the assessment, while the absence of rechallenge and the fictional nature of the case limit certainty.

Bradford Hill considerations vs patient-level causality

The task material notes that Bradford Hill considerations are intended as a framework for causal interpretation, particularly in epidemiological or population-level evidence. They are not a simple patient-level scoring algorithm. For this case, the most visible consideration is temporality; the project therefore avoids presenting Bradford Hill as a numerical patient score.

4. Safety Context and Reporting

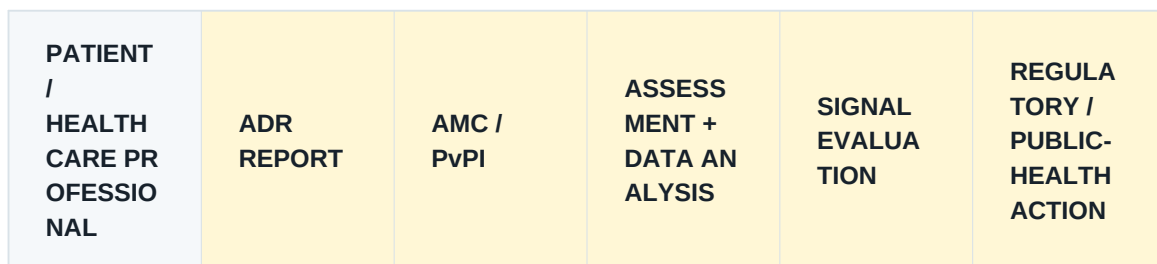
Pregnancy-risk terminology and ADR reporting were included in the training brief. The historical FDA A-X letter system is useful for understanding older references, while current FDA prescription labeling uses narrative pregnancy-risk information instead.

Historical category	Training-level meaning
A	Adequate controlled studies did not demonstrate fetal risk.
B	Animal data did not show fetal risk, while adequate human evidence was lacking or not confirmatory.
C	Animal studies showed adverse fetal effect or evidence was insufficient; benefit-risk consideration was required.
D	Human fetal risk evidence existed, but benefits could warrant use in some circumstances.
X	Fetal risk outweighed possible benefit; use in pregnancy was contraindicated.

Core fields for an ADR report in India

No.	Information to capture
01	Patient initials / demographic information
02	Reaction onset date and reaction description
03	Management, outcome and seriousness criteria where applicable
04	Suspected medicine(s) - dose, route, frequency and therapy dates
05	Action taken with suspected medicine and rechallenge information, if applicable
06	Concomitant medicines / products and relevant investigations
07	Reporter details

Simplified reporting pathway



The task material describes PvPI as India's national drug-safety monitoring programme and notes that its National Coordination Centre is located at the Indian Pharmacopoeia Commission (IPC), Ghaziabad.

5. Project Interpretation and Key Takeaways

- 1. Start with the event.** Define what happened before deciding whether a medicine caused it.
- 2. Build a timeline.** A clear exposure-to-onset sequence is central to basic ADR identification.
- 3. Review all medicines.** New, long-standing and concomitant therapies should initially be considered.
- 4. Look for dechallenge information.** Improvement after withdrawal can support the suspected relationship.
- 5. Do not overstate causality.** A temporal link plus dechallenge may support a preliminary assessment, but does not automatically establish definitive causality.
- 6. Document missing evidence.** A case is more credible when it explicitly states whether rechallenge, investigations or alternative-cause information are available.
- 7. Reporting can occur before certainty.** The task material emphasizes that suspected ADRs can be reported even when causality has not been definitively established.

Quality checklist used in this project

Check	Status	Evidence in report
Fictional educational case clearly labelled	Complete	Cover page + case section
AE vs ADR distinction	Complete	Foundations section
ADR Type A-F overview	Complete	ADR pattern table
Timeline + suspected medicine	Complete	Case timeline
Temporal relationship / dechallenge / rechallenge	Complete	Causality reasoning
Alternative causes considered	Complete	Assessment matrix
India reporting pathway	Complete	Reporting section
Avoids copying source case wording	Complete	Original fictional scenario

Submission-ready note: This PDF is designed to sit inside a GitHub repository as the main visual deliverable. The accompanying README should explain the case, file structure and learning objective without reproducing another student's work.

6. Source Basis and Declaration

This project was developed from the internship Task 1 material supplied in the conversation. The source material covers pharmacovigilance basics, AE vs ADR, Type A-F classification, a fictional ADR-identification scenario, pregnancy-risk terminology, causality approaches (Bradford Hill, WHO-UMC and Naranjo), ADR reporting in India, PvPI, and the public-health importance of reporting. The present submission preserves that learning scope while using a distinct case scenario and original wording.

Important scope note

This is an educational exercise. It is not a real patient case, a medical record, or a substitute for clinical assessment. Any real-world ADR evaluation must use complete patient information and the applicable reporting and clinical procedures.

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END OF TASK 1 PROJECT

Prepared as a structured, original educational submission.